GROUP NO. 3 **Date: August 25, 2026**

Scenario Title: Scenario B (Higher Education) — Integrating an Online Student Registration Portal with the University Cashiering Database.

Sub-Team	Members Assigned	Responsibilities
A — REST/JSON Engineers	Charls Sampaga [One Member]	Design, format, and validate the REST JSON registration request payload.
B — Legacy SOAP/XML Engineers	Mhar Granado Messiahlyn Mae Mabilangan	Design, format, and validate the equivalent SOAP XML backend transaction.
C — Data Transformation Specialists	Stephanie April Romo [One Member] [One Member]	Map Student Registration Database (MySQL) fields to the target JSON payload keys and define transformation rules.
D — Middleware Testers & QA	Mike Casaverde Lanz Michael Casino	Configure Postman, execute API requests, verify headers/status codes, assemble final reports.

Task 1: Payload Construction

- **REST JSON Payload (Sub-Team A):**


```
{
  "registrationId": "REG-2026-00001",
  "studentId": "202301095",
  "fullName": "Pagkaliwagan, Kian Lhei",
  "program": "Bachelor of Science in Information and Technology",
  "semester": "1st Semester",
  "academicYear": "2026-2027",
  "registrationDate": "2026-08-25",
  "courses": [
    {
      "courseCode": "ITEC116",
      "courseTitle": "System Integration and Architecture 2",
      "units": 3,
      "schedule": "W 7:00-9:00, 15:00-17:00"
    },
    {
      "courseCode": "ITEC11A",
      "courseTitle": "Integrated Programming and Technologies 2",
      "units": 3,
      "schedule": "W 7:00-9:00, W 15:00-17:00"
    }
  ],
  "totalUnits": 8,
  "totalHour": 7,
  "assessment": {
    "totalAmount": 9345.00,
    "amountPaid": 2000,
```

```
    "outstandingBalance": 7345.00,
    "paymentStatus": "Partial"
  },
  "accountStatus": true
}
```

Task 2: Data Transformation & Schema Mapping

```
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
   <soap:Header/>
   <soap:Body>
      <EnrollmentRequest id="REG-2026-00001">
         <Student id="202301095" active="true">
            <StudentId>202301095</StudentId>
            <FullName>Pagkaliwagan, Kian Lhei</FullName>
            <AccountStatus>true</AccountStatus>
         </Student>
         <Course>
            <Program>Bachelor of Science in Information Technology</Program>
         </Course>
         <Enrollment>
            <Semester>1st Semester</Semester>
            <AcademicYear>2026-2027</AcademicYear>
            <RegistrationDate>2026-08-25</RegistrationDate>
            <Status>Enrolled</Status>
            <Courses>
               <Course>
                  <CourseCode>ITEC116</CourseCode>
                  <CourseTitle>System Integration and Architecture 2</CourseTitle>
                  <Units>3</Units>
                  <Schedule>W 7:00-9:00, 15:00-17:00</Schedule>
               </Course>
               <Course>
                  <CourseCode>ITEC11A</CourseCode>
                  <CourseTitle>Integrated Programming and Technologies 2</CourseTitle>
                  <Units>3</Units>
                  <Schedule>W 7:00-9:00, 15:00-17:00</Schedule>
               </Course>
            </Courses>
            <TotalUnits>6</TotalUnits>
            <TotalHours>7</TotalHours>
         </Enrollment>
         <Assessment>
            <TotalAmount>9345.00</TotalAmount>
            <AmountPaid>2000.00</AmountPaid>
            <OutstandingBalance>7345.00</OutstandingBalance>
            <PaymentStatus>Partial</PaymentStatus>
         </Assessment>
      </EnrollmentRequest>
   </soap:Body>
</soap:Envelope>
```

Task 3: API Endpoint Execution & Validation (Sub-Team D)

Body:

Headers:

Group Laboratory Activity 2: Enterprise Payload Design & Schema Transformation
Course Code / Title: ITEC 116 – System Integration and Architecture
Prerequisite: Group Activity 1 (System Integration Blueprinting)

Deliverable: 1 Group Laboratory Report (PDF) submitted per team

Team Sub-Groups & Role Allocations (8–9 Members)
To ensure every student actively contributes, split your group into four sub-teams:

- **Sub-Team A: REST/JSON Engineers (2 Members)**
  - *Members Assigned:* Student 1 & Student 2
  - *Responsibilities:* Design, format, and validate the modern REST JSON request payload.
- **Sub-Team B: Legacy SOAP/XML Engineers (2 Members)**
  - *Members Assigned:* Student 3 & Student 4
  - *Responsibilities:* Design, format, and validate the equivalent legacy SOAP XML schema.
- **Sub-Team C: Data Transformation Specialists (2–3 Members)**
  - *Members Assigned:* Student 5, Student 6, (and Student 7 if 9 members)
  - *Responsibilities:* Map source database fields to target payload keys and define data transformation rules.
- **Sub-Team D: Middleware Testers & QA (2 Members)**
  - *Members Assigned:* Student 8 & Student 9
  - *Responsibilities:* Configure Postman, execute API requests, verify HTTP headers/status codes, and assemble the final report.

Step-by-Step Group Tasks
Task 1: Payload Construction (Sub-Teams A & B)
Using the enterprise scenario established in Activity 1 (Retail, Education, or Healthcare), construct two equivalent data payloads representing a single transaction (e.g., submitting an order, enrolling in a course, or booking a medical appointment):

- **REST JSON Payload (Sub-Team A):** Draft a clean JSON object containing nested arrays (e.g., ordered items, student course lists, or patient medical histories).
- **SOAP XML Payload (Sub-Team B):** Draft an equivalent XML schema using structured tags, attributes, and WSDL-compliant formatting.

Task 2: Data Transformation & Schema Mapping (Sub-Team C)
Map raw backend database fields (from Activity 1) to the target JSON API payload created by Sub-Team A. Complete the following matrix:

Legacy Backend DB Field	Legacy Data Type	Target API Key (JSON)	Target Data Type	Transformation / Business Rule
stud_no	VARCHAR(12)	studentId	String	Trim whitespace; convert hyphens.
f_name, l_name	VARCHAR(50)	fullName	String	Concatenate (l_name + ", " + f_name).
is_active	INT(1)	accountStatus	Boolean	Convert 1 to true, 0 to false.

Task 3: API Endpoint Execution & Validation (Sub-Team D)
Import the JSON payload into **Postman** and test it against a mock REST endpoint (e.g., reqres.in/api/users or jsonplaceholder.typicode.com/posts):

1. Configure a POST request with HTTP Header Content-Type: application/json.
2. Execute the request and verify a successful **HTTP 201 Created** or **200 OK** response.

3. Capture a high-resolution screenshot displaying the Request Body, Request Headers, Status Code, and Response Payload.
4.

Final Deliverable Checklist (1 PDF per Group)

Your submitted report must follow this layout:

1. **Team Roster:** Group Number, Scenario Title, and Name List broken down by Sub-Teams A, B, C, and D.
2. **Section 1 (Task 1):** Validated JSON and XML code blocks.
3. **Section 2 (Task 2):** Completed Data Field Mapping & Transformation Matrix.
4. **Section 3 (Task 3):** Postman execution screenshot proof.

Evaluation Rubric (50 Points Total)

- **JSON & XML Payload Accuracy (15 pts):** Payloads are syntactically sound and accurately reflect the enterprise scenario.
- **Schema Mapping & Transformation Logic (15 pts):** Data casting, string concatenation, and value mapping rules are realistic.
- **Postman API Test Proof (10 pts):** Screenshot confirms valid request execution, correct headers, and status code verification.
- **Sub-Team Organization & Role Execution (10 pts):** All 8–9 members are clearly listed with assigned tasks, and deliverables are cleanly integrated.